

Module 2

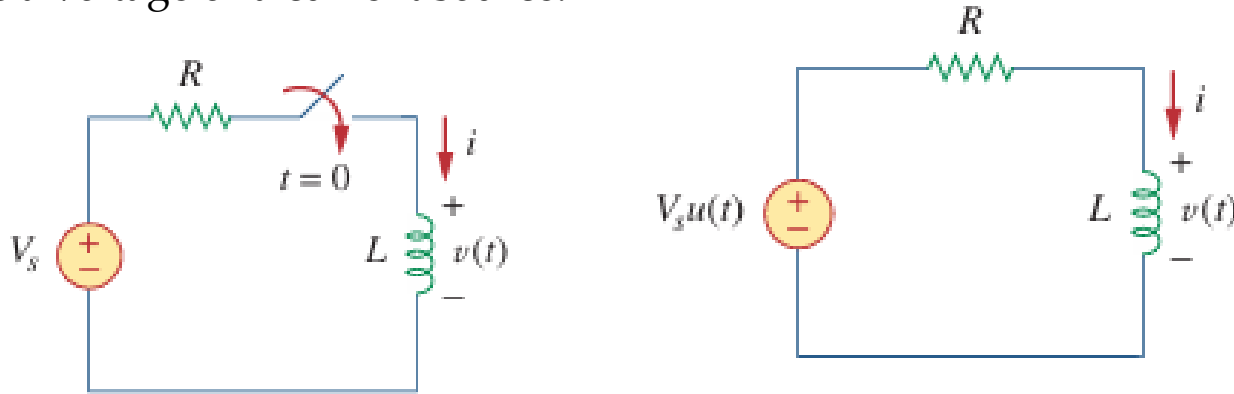
Transient Analysis of First and Second-Order Circuits

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Step Response of RL circuit

The step response of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.



Let the response be the sum of the transient response and the steady-state response.

$$i = i_t + i_{ss}$$

$$i_t = Ae^{-t/\tau}, \quad \tau = \frac{L}{R} \quad \text{where } A \text{ is a constant to be determined}$$

The steady-state response is the value of the current a long time after the switch is closed. At that time, the inductor becomes a short circuit, and the voltage across it is zero. The entire source voltage V_s appears across R .

$$i_{ss} = \frac{V_s}{R}$$

Step Response of RL circuit

$$i = Ae^{-t/\tau} + \frac{V_s}{R}$$

We now determine the constant A from the initial value of i. Let I_0 be the initial current through the inductor, which may come from a source other than $I_0 V_s$.

$$i(0^+) = i(0^-) = I_0$$

Thus, at $t = 0$

$$I_0 = A + \frac{V_s}{R}$$

$$A = I_0 - \frac{V_s}{R}$$

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-t/\tau}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$

Step Response of RC circuit

If the switching takes place at time $t = t_0$ instead of $t = 0$

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

If $I_0 = 0$,

$$i(t) = \begin{cases} 0, & t < 0 \\ \frac{V_s}{R}(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

$$i(t) = \frac{V_s}{R}(1 - e^{-t/\tau})u(t)$$

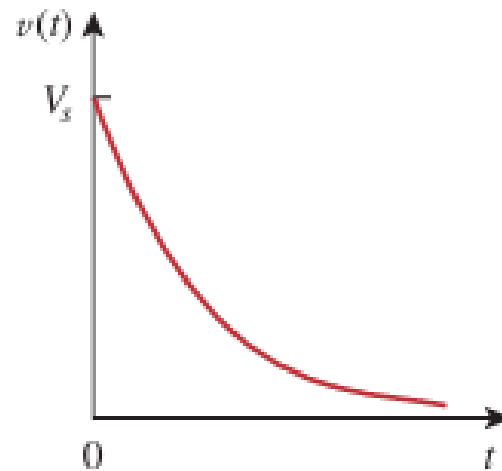
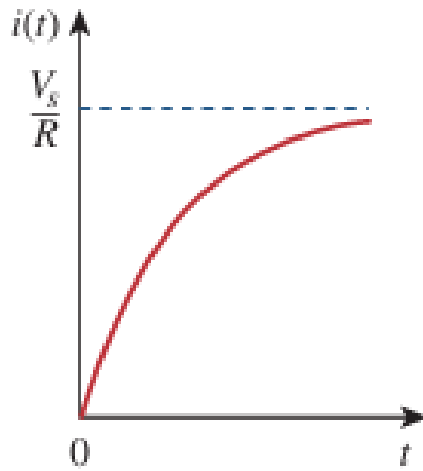
The voltage across the inductor

$$v(t) = L \frac{di}{dt} = V_s \frac{L}{\tau R} e^{-t/\tau}, \quad \tau = \frac{L}{R}, \quad t > 0$$

$$v(t) = V_s e^{-t/\tau} u(t)$$

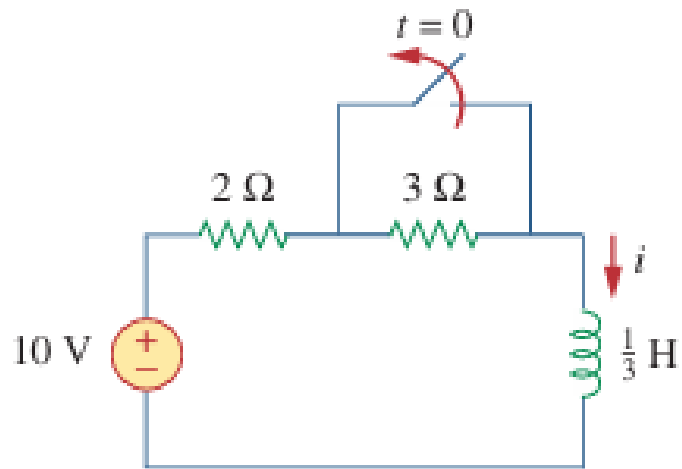
Step Response of RC circuit

the step responses are as shown below.



Exercise

1. Find $i(t)$ in the circuit for $t > 0$. Assume that the switch has been closed for a long time.



Next class: Source Free RLC circuit